Tone Analysis
in
AI Interview

By

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Faculty of Computing and Information Technology
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Declaration

The project submitted herewith is a result of my own efforts in totality and in every aspect of the project works. All information that has been obtained from other sources had been fully acknowledged. I understand that any plagiarism, cheating or collusion or any sorts constitutes a breach of TAR University rules and regulations and would be subjected to disciplinary actions.

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Abstract

This project focuses on analyzing emotions and personality traits of candidates from asynchronous video interviews (AVIs), addressing key challenges in traditional recruitment processes by enhancing objectivity and efficiency in candidate evaluation. By leveraging audio-visual data, the system identifies non-verbal cues and behavioral patterns at the sentence level, providing recruiters with deeper insights into candidate performance.

The methodology involves advanced audio feature extraction using Praat and Librosa, focusing on prosodic, spectral, and voice quality features. Statistical summaries, including mean, minimum, maximum, and standard deviation, are computed for low-level features extracted from the audio. Machine learning models such as Recurrent Neural Networks (RNN), Support Vector Machines (SVM), XGBoost, and Random Forest are employed for emotion recognition and personality inference. These models are trained on benchmark datasets such as Chalearn First Impressions V2 for personality perception and RAVDESS for speech emotion recognition, ensuring robust performance. Real-world interview data is incorporated to evaluate the system under deployment conditions.

The system processes audio data to extract features that are standardized and aggregated for analysis. Models are trained to identify emotions and infer Big Five personality traits, addressing challenges such as overlapping traits across time windows. Classification performance is evaluated using metrics like accuracy, precision, recall and F1-Score metrics, demonstrating the system's scalability and reliability.

Initial results show the system's effectiveness in accurately classifying emotions and inferring personality traits, contributing to enhanced decision-making in recruitment. Despite its successes, challenges such as refining aggregation techniques and addressing inconsistencies in detecting overlapping traits persist. This project underscores the value of integrating advanced emotion and personality analytics into recruitment processes, offering a scalable, data-driven solution to improve candidate evaluations while setting a foundation for further research and optimization.

Acknowledgement

First of all, I would like to express my deepest gratitude to my supervisor, Mr. Kong Yih Hern, for giving me this invaluable opportunity to undertake this project and for providing tremendous support throughout the entire process. His clear guidance and ability to point out my mistakes have been instrumental in helping me improve and navigate the complexities of this project. Mr. Kong's constructive feedback and unwavering support have provided me with the right direction and methodology to successfully carry out this work. It was truly a privilege and pleasure to work under his guidance, and I deeply appreciate his patience, dedication, and encouragement throughout this journey. This project would not have been possible without his invaluable insights and assistance. Thank you very much, Mr. Kong, for your tremendous support and guidance.

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Chapter 1

Introduction

1 Introduction

This project is motivated by the needs of GIT, a recruitment company in Indonesia, which seeks a more robust and customized approach to the recruitment process. GIT oversees the entire hiring process, from sourcing and screening candidates to conducting interviews and negotiating job offers. Traditional interview methods, rely on face-to-face interactions are often time consuming and susceptible to biases, particularly in Indonesia’s competitive job market. To enhance efficiency and objectivity in the hiring process, this project explores the use of Asynchronous Video Interviews (AVIs). AVIs enable candidates to record responses to predefined questions without real-time interaction with interviewers. AVI is deployed with artificial intelligence algorithms to automatically identify the candidate's personality traits or other variables of interest. This project will focus on deploying the tone analysis algorithm used in AVIs to provide deeper insights into candidates' emotional and personality traits. Tone analysis involves extracting audio features from recorded responses and applying algorithms for Speech Emotion Recognition (SER) and Automatic Personality Perception (APP). SER identifies emotional states from speech, while APP assesses perceived personality traits based on observed speech patterns. These advanced techniques aim to move beyond traditional assessments, offering a more advanced understanding of candidates, ultimately enhancing the recruitment process for GIT.

The problem faced by GIT with their current recruitment process:

1. Inefficient in a large number of candidate evaluations. The manual review and evaluation of interview data is resource intensive and time consuming, particularly when dealing with a large number of candidates in the Indonesian market. This inefficiency can delay the hiring process and reduce the overall effectiveness of candidate assessment, especially in a competitive job market where timely decisions are important.
2. Poor emotion detection while interviewing. Traditional interview methods often fail to accurately capture the details of a candidate’s emotional state due to relying on subjective human interpretation and limited analytical tools. This can lead to inconsistent or incomplete understanding of candidates' feelings about the interview questions and topics.
3. Insufficient insight into Personality Traits from candidate speech. Traditional interview processes often lack detailed analysis of personality traits, which are crucial in determining how well an applicant fits for a given role and the culture of the organization.

1.1 Objectives

To solve the problem mentioned, this project aimed to design a tone analysis algorithm or methodology that is able to deploy into the GIT's AVI system for enhancing the objectivity of the evaluation process and provide a more comprehensive understanding of a candidate's emotions and personality traits in speech.

1.1.1 Speech emotion recognition (SER) model

The SER model focuses specifically on identifying emotions of candidates from the speech. This can be achieved by analyzing various speech features (e.g., MFCCs, pitch, energy) and mapping them to specific emotions such as happiness, sadness, anger, or fear. Machine learning algorithms are used to train the SER model on the extracted features. Common algorithms include deep learning models such as Convolutional Neural Networks (CNNs) or Recurrent Neural Networks (RNNs) and ensemble models like Random Forests or Gradient Boosting Machines (GBMs). By accurately detecting emotions, the SER model provides insights into a candidate's emotional state during the interview. This can reveal how a candidate feels about specific questions or topics, which can be crucial for understanding their real reactions and attitudes.

1.1.2 Automatic Personality Perception (APP) model

The APP model aims to use audio features extracted from speech to predict personality traits based on the Big Five personality framework (Openness, Conscientiousness, Extraversion, Agreeableness, Neuroticism), which is is commonly used to predict candidate suitability in a specific job role and company cultural environment. This model will also be trained using machine learning algorithms including deep learning models such as Convolutional Neural Networks (CNNs) or Recurrent Neural Networks (RNNs) and ensemble models like Random Forests or Gradient Boosting Machines (GBMs). By predicting personality traits from speech, the APP model provides a deeper understanding of a candidate's character and suitability for specific roles. This insight can complement traditional resume assessments and interviews, offering a more holistic view of the candidate.

1.1.3 Enhance Feature Representation

The goal here is to refine and optimize the feature set used for both emotion and personality classification by experimenting with various acoustic (pitch, energy, zero crossing rating) and prosodic (MFCC, LPC) features. This involves selecting the most effective features for capturing both emotional nuances and personality indicators in speech. By enhancing feature representation to capture both emotional and personality cues, the algorithm can provide a more reliable and accurate analysis of both emotions and personality traits. This dual focus ensures that the AVI system captures a wide range of speech characteristics relevant to understanding candidates comprehensively.

1.1.4 Identify the Most Suitable Algorithm for Training Models

Develop and assess various machine learning algorithms to determine the most effective approach for training both the Speech Emotion Recognition (SER) and Automatic Personality Perception (APP) models. This includes evaluating ensemble models, neural network architectures, and other advanced techniques to enhance model performance. By experimenting both Ensemble Models Techniques like Random Forests, Gradient Boosting Machines (GBMs), or stacking methods that combine multiple models to improve prediction accuracy and robustness and Neural Networks Architectures like Convolutional Neural Networks (CNNs) or Recurrent Neural Networks (RNNs), the most suitable or better performance model is chosen for deployment in SER and APP. This ensures that the models are not only accurate but also robust and scalable, providing reliable insights into candidates' emotions and personalities. By testing various algorithms, the project can find the best fit for the specific characteristics of the speech data and improve overall model effectiveness.

1.1.5 Develop a Robust Data Preprocessing Framework

Create a standardized data preprocessing pipeline to handle audio data from interviews. This includes steps such as noise reduction, normalization, segmentation, and feature extraction to ensure high-quality input for emotion and personality models. Developing a robust data preprocessing framework is essential for maintaining the accuracy and reliability of SER and APP models. By ensuring that both training and deployment data are processed using the same techniques, the model can accurately predict emotions and personality traits from new audio data. Models learn to recognize patterns and relationships in the data that is completely preprocessed. When new data in AVI is processed in the same way, the model can accurately apply the patterns it learned during training to make predictions. If preprocessing steps differ, the model might not correctly interpret the new data, leading to inaccurate predictions. Consistent preprocessing not only enhances the model's performance but also ensures fair and reliable assessments during real-world deployment.

1.2 Background

1.2.1 GIT

This project is motivated by the needs of GIT, a recruitment company in Indonesia, which seeks a more robust and customized approach to the recruitment process. GIT is a recruitment agency that manages the entire hiring process for businesses, from finding and screening candidates to conducting interviews and negotiating job offers. The process of the recruitment agency company included collaborating with businesses to identify and recruit candidates for job vacancies, creating job descriptions, reviewing applications, and conducting initial interviews. The final hiring decisions are typically made by the company based on the recommendation provided by the recruitment agency. Employment interviews are one of the most frequently used methods for personnel selection in recruitment agencies (Tschöpe, 2020). Traditional interviewing has relied on face to face interaction. The selection process might involve a resume document review, followed by an invitation, meeting with the interviewer for the interview process. Then the candidate's responses will be evaluated by the interviewers based on their communication skill and performance. This approach can be time-consuming, resource-intensive, and easily affected by human biases, which delays the efficiency and effectiveness of candidate tests (Kazim, 2024). In a country like Indonesia, with its large population and highly competitive job market, these inefficiencies become even more apparent. To enhance efficiency and objectivity in the hiring process, this project explores the use of Asynchronous Video Interviews (AVIs).

1.2.2 Asynchronous video interviews (AVIs)

Asynchronous video interviews (AVIs) is an online interview technology for personnel selection. Importantly, in an AVI there is no synchronous, back and forth communication, and no interviewer to interact with. The data collected from candidates in an AVI are similar to those collected in two‑way interviews, but the interviewer is absent, candidates do not have to interact directly with another human by asking questions. An AVI allows candidates to log into the platform and record the response to predefined interview questions via a device equipped with a camera and microphone (Roulin, 2020). Ultimately, applicants' recorded responses will be scored by an interviewer or automatically scored by an AI computer algorithm. Traditionally, responses of the candidates are recorded and stored for later evaluation by HR employers (Patrick D. Dunlop, 2021). However, to speed up the selection process, artificial intelligence algorithms are deployed to automatically identify the candidates personality traits or other variables of interest (Antonis Koutsoumpis, 2024). One of the useful algorithms used in the automatic approach is tone analysis as language can be a useful window that enables us to understand the social and psychological attributes of its users, including personality traits (Lee, Sinae & Park, 2021). Other artificial intelligence algorithms included sentiment analysis, facial micro/macro expression detection, deception detection, and body gesture. These approaches are able to identify the candidates' personality traits, abilities and skills and emotional states (H. -Y. Suen, 2019)

(Rasipuram, 2016). Those algorithms can be trained by using machine learning techniques from participants' automatically extracted audio (voice characteristics), visual (facial expressions), and verbal (text) features from the recorded response video.

1.2.3 Tone Analysis in AVI

Tone Analysis in AVI not only enhances the objectivity of the evaluation process but also contributes to a more comprehensive understanding of a candidate's emotions and personality traits. Tone analysis is used in AVI to analyze the speech of the candidates. It is used by extracting the audio feature from the video and performing speech emotion recognition (SER) and Automatic Personality Perception (APP) by using the extracted audio feature. SER is a method that involves recognizing the emotional aspects of speech irrespective of the semantic content (Obrist & Velasco, 2020). A typical SER system can be considered as a collection of methodologies that isolate, extract and classify speech signals to detect emotions embedded in them (Mehmet Berkehan Akçay & Kaya Oğuz, 2020). APR is used for automatically mapping prosodic aspects of speech into personality traits attributed by human listeners based on their observed behavior (Rangra et al., 2023). Unlike Automatic Personality Recogni-tion (APR), which identifies a person's true personality from behavioral evidence (Rangra et al., 2023), the goal of APP is not to predict the real personality of an individual, but the personality as perceived by observers. In other words, APP is not expected to predict the real personality of a given person, but the personality that others attribute to her in a given situation (Gelareh Mohammad & Alessandro Vinciarelli, 2012). After extracting out the audio features such as mean pitch, pitch variation and speaking rate from the responded video, the tone analysis algorithm is able to classify the emotion such as anger or happiness and personality traits such as Openness, Conscientiousness, Extroversion, Agreeableness, and Neuroticism (Big Five Personality Traits) of the person. By incorporating advanced tone analysis, the hiring process can move beyond superficial assessments to reveal deeper psychological and emotional insights that are critical to making informed hiring decisions.

1.2.4 Big Five Personality Traits

The Big Five Personality Traits (BF), also called the five-factor model (FFM) or OCEAN model, is a model that have been identified by applying factor analysis to the large number of words describing personality in everyday language (Gelareh Mohammad & Alessandro Vinciarelli, 2012). This widely examined theory suggests five broad dimensions used by some psychologists to describe the human personality and psyche (Zhang, 2017). The five dimensions are as follows:

1. Openness: appreciation for art, emotion, adventure, unusual ideas, original, wide interests, etc.

2. Conscientiousness: a tendency to be organized and dependable, show self-discipline, act dutifully, aim for achievement, and prefer planned rather than spontaneous behavior, etc.

3. Extroversion: active, energy, positive emotions, assertiveness, sociability assertive, etc.
4. Agreeableness: a tendency to be compassionate and cooperative rather than suspicious and antagonistic towards others, appreciative, kind, generous, forgiving, sympathetic, trusting, etc.

5. Neuroticism: The tendency to experience unpleasant emotions easily, anxious, self-pitying, tense, touchy, unstable, worrying, etc.

This construct is commonly used to predict whether a job candidate will perform well in a specific job role and engage well in a prospective cultural environment (H. -Y. Suen, 2019). Although a variety of models can be used to assess personality, BF provides researchers and practitioners with a well-defined taxonomy for selecting job applicants.

1.3 Advantages and Contributions

The purpose of this tone analysis algorithm is to help our project clients, GIT, to handle the audio related features from the video of the candidate's recorded response in their AVI system. The main advantages are as follows:

1.3.1 Provide deeper insights into candidates' emotional and personality traits

By analyzing candidates' emotional tones as well as personality traits, the tone analysis algorithm provides a deeper level of insight of the candidate. This deeper insight enables GIT to identify key characteristics such as emotional intelligence, and cultural fit of the candidate. The algorithm's ability to extract the audio feature allows for the detection of traits that might not be immediately shown in a traditional interview. By integrating these insights into GIT's broader recruitment strategies, the company can tailor its approach to candidate sourcing and job matching. For instance, understanding a candidate's emotional responses can help predict how they might handle stressful situations or work in team environments. Additionally, personality traits from the analysis can also help to identify long-term role suitability and career development of the candidate. This approach makes GIT's recruitment process more effective by ensuring that candidates not only meet the necessary qualifications but also fit well with the company's culture and job requirements.

1.3.2 Enhance Consistency in Candidate Evaluation

One of the significant challenges in traditional interview processes is the variability present in human evaluators. Different interviewers might interpret a candidate's responses differently, leading to inconsistencies and potential biases in the evaluation process. The tone analysis algorithm solves this issue by providing a standardized framework to evaluate candidates. The algorithm evaluates all candidates based on the same set of criteria, analyzing their speech for specific emotional and personality markers. This consistency ensures that every candidate gets evaluated with the same standards, reducing the influence of subjective interpretation and personal biases. As a result, GIT can achieve a higher degree of fairness and transparency in its hiring process. Consistent evaluations not only improve the quality of hire but also build trust in the recruitment process.

1.3.3 Speed up the selection process

The recruitment process can often be time-consuming, particularly when dealing with a large pool of candidates. By integrating with the AVI system, which automates the evaluation of candidates' speech in their recorded responses, the tone analysis algorithm helps to speed up this process. The algorithm processes and analyzes responses automatically, providing fast and accurate evaluations, compared to manually reviewing and interpreting each interview. This efficiency allows GIT to handle a high volume of candidate responses without losing the quality of its assessments. The company can efficiently adjust its hiring efforts by quickly evaluating applicants and identifying those who fit the desired emotional and personality profiles. This is especially helpful for filling positions that require a quick turnaround or during periods of high demand, such as seasonal hiring. GIT can improve overall recruitment efficiency and ensure that top candidates are found before they are grabbed up by competitors by speeding up the selection process and reducing the time-to-hire.

1.4 Project Plan

Project Schedule:

ACTIVITIES	EXPECTED OUTCOME	COMPLETION DATE
Proposal Development	Define the problem and proposed solution	8/8/2024
Chapter 1 Introduction	Comprehensive overview of the project scope and objectives	29/8/2024
Chapter 2 Research Background	Detailed literature review and identification of research gaps	12/9/2024
Chapter 3 Methodology and Requirements Analysis	Detailed methodology and requirements analysis report	29/9/2024
Chapter 4 System Design	Complete system architecture and design specifications	28/11/2024
Final Project 1 Report	Document covering all work and findings of the first project phase	1/12/2024
Data Analysis	Understand all the data	13/12/2024
Preprocessing and Feature Engineering	Cleaned and transformed dataset ready for model training	28/12/2024
Model Selection and Training	Selection of optimal models and initial training results	10/1/2025
Model Testing and Validation	Validation reports with metrics showcasing model accuracy and reliability	24/1/2025
Deployment	Successful deploy the system	10/2/2025
System Preview by Supervisor	Approval and feedback from supervisor after system demonstration	23/2/2025
System Enhancement	Implementation of improvements and refinements based on feedback	8/3/2025
Final System Test	Final testing results demonstrating system performance and stability	9/3/2025
Chapter 5 : Implementation & Testing	Detailed documentation of system implementation and testing processes	18/3/2025
Chapter 6 : Deployment	Documentation on the deployment strategy and execution	30/3/2025
Chapter 7 : Conclusion	Summary of findings, outcomes, and project impact	6/4/2025
Final FYP Report	Comprehensive final report covering all aspects of the project	20/4/2025

1.5 Project team and Organization

This project is a collaborative effort, organized into specialized roles to effectively develop the AI algorithm for the Asynchronous Video Interview (AVI) system. The team is structured as follows:

- **Tone Analysis (audio):** This component is handled by Choong Zhi Yang and Chong Kit Yan, focuses on analyzing the tone of candidates' speech to extract meaningful emotional and personality-related features. This analysis will provide deeper insights into candidates' suitability for various roles.
- **Facial Expression Analysis (visual):** Chai Zhi Xiang is responsible for developing the algorithms that will detect and interpret candidates' facial expressions during interviews.
- **Text Relevance Analysis (text):** Dorain Lim Kai Zhun will handle the text relevance analysis, focusing on the sentiment analysis, personality traits, and deception detection using text that is translated from audio.

1.6 Chapter Summary and Evaluation

This chapter outlines the objectives and structure of the project aimed at enhancing GIT's recruitment process through the integration of Asynchronous Video Interviews (AVIs) and advanced tone analysis algorithms. The primary challenge identified is the inefficiency and bias associated with traditional face-to-face interviews, especially in the context of Indonesia's competitive job market. To address these issues, the project focuses on deploying a tone analysis algorithm that provides deeper insights into candidates' emotional and personality traits. This includes the development of Speech Emotion Recognition (SER) and Automatic Personality Perception (APP) models, which analyze speech features to assess emotions and predict personality traits based on the Big Five framework. The project also emphasizes the importance of refining feature representation and selecting the most suitable machine learning algorithms, including ensemble models and neural networks, to enhance the accuracy and reliability of the analysis. By integrating these advanced techniques into the AVI system, the project aims to not only speed up the selection process but also provide a more consistent and objective evaluation of candidates. The tone analysis algorithm will help GIT gain a comprehensive understanding of candidates' emotional states and personality traits, ultimately improving the effectiveness of their recruitment strategy. The approach promises to address inefficiencies and biases, leading to more informed hiring decisions and a more efficient recruitment process.

Chapter 2

Literature Review

2 Literature Review

This chapter explores the most important elements of developing successful algorithms for Speech Emotion Recognition (SER) and Automatic Personality Prediction (APP) in the context of Asynchronous Video Interviews (AVIs). It covers four key areas essential to the model development process: Data Preprocessing, Audio Feature Extraction and Selection, Emotion and Personality Labeling, and Model Selection.

The Data Preprocessing section discusses techniques such as framing, windowing, and denoising, which are crucial for preparing raw audio signals for analysis. The Audio Feature section explores various approaches to extracting and selecting relevant features from speech data, including traditional acoustic features and more recent deep learning-based methods. The Emotion and Personality Labeling section investigates different frameworks for categorizing emotions and personality traits, with a focus on their suitability to the interview context. Finally, the Model Selection section reviews various machine learning algorithms and architectures that have shown promise in SER and APP tasks, comparing their performance and suitability for AVI analysis.

2.1 Data Preprocessing

Data preprocessing is a critical step in audio-based emotion and personality detection, ensuring that the raw audio signals are transformed into usable formats for machine learning algorithms. Various techniques, including framing, windowing, and denoising, have been employed in previous research to address challenges posed by non-stationary audio signals, noisy environments, and the need for standardized data input.

2.1.1 Framing and windowing

Signal framing, also known as speech segmentation, is the process of partitioning continuous speech signals into fixed length segments to overcome several challenges in audio signal processing. It addresses the non-stationary nature of speech signals and enables the extraction of meaningful features (Madanian et al., 2020). Frameshift / overlapping is a key concept in framing, which describes overlapping portions between current and previous frames. The frame size and the overlap ratio are determined during the pre-processing to help differentiate the starting points of each frame and the duration of the frames (Basu et al., 2017). The overlap rate is employed to reduce information loss and soften the transition from one frame to the next (Kamil, 2019).

Emotion can change in the course of speech since the signals are non-stationary. However, speech remains invariant for a sufficiently short period, such as 10 to 30 ms. By framing the speech signal, this quasi-stationary state can be approximated, and local features can be obtained. Additionally, the relation and information between the frames can be retained by deliberately overlapping 30% to 50% of these segments. Continuous speech signals restrain the usage of processing techniques such as Discrete Fourier Transform (DFT) for feature extraction in applications such as SER. Consequently, fixed size frames are suitable for classifiers, such as Artificial Neural Networks, while retaining the emotion information in speech (Madanian et al., 2020).

Recent studies in emotion recognition and personality prediction have employed various window sizes and step sizes. In a study using NPSO architecture in deep learning, researchers normalized and resampled data to a frame of 3 seconds for each voice sample across four different datasets. Notably, they intentionally added noise to enhance the model's ability to distinguish between various emotions and personalities (Rangra et al., 2023).

For psychometric analysis of AVIs, researchers computed audio features using a sliding window of 25 ms with a step size of 10 ms to traverse the entire video (Koutsoumpis et al., 2024) . In a study on automatic personality assessment, researchers used a time step of 15 ms and a window length of 6ms (Batrinca et al., 2011). In (Mohammadi et al., 2010) mapping nonverbal vocal behavior into trait attribution, researchers extracted low-level features using PRAAT (version 5.1.15) from 40 ms windows at regular time steps of 10 ms. A study on the efficiency of speech descriptors in relation to emotion recognition set the analysis window to 20 ms with 50% overlap (Kamińska & Sapiński, 2017).

The variation in window sizes and step sizes across these studies suggests that these parameters may need to be optimized based on the specific task, dataset, and research objectives. Researchers should consider experimenting with different window sizes and step sizes to determine the most effective configuration for their particular application in SER or personality prediction.

2.1.2 Denoising

Denoising is a crucial preprocessing step in audio signal processing, particularly for speech emotion recognition (SER) and personality prediction tasks. Various denoising methods have been investigated and compared in recent literature, each with its own strengths and applications.

(Pohjalainen et al., 2016) investigated audio signal denoising methods in cepstral and log-spectral domains, comparing them with common implementations of standard techniques. The audio signal is processed in Hann-windowed frames of 30 ms extracted every 10 ms. It found that denoising has improved the results in a majority of the evaluated training/test setups and in these cases, the proposed methods outperform the standard baselines. This approach provides a comprehensive view of denoising effectiveness across different signal representations. Therefore, it is noted that the proposed methods show promise in audio noise reduction an audio content analysis

While ZCR and STE are conventional methods for speech processing, more recent studies have shown that newer techniques often outperform these traditional approaches. For instance, a study by Bachu et al. (2008) proposed a method that performed better in experiments compared to the conventional ZCR & STE method. The method uses statistical properties of background noise and also the physiological aspects of the speech production process. The method assumes the noise to be white Gaussian.

The MMSE approach, particularly for short-time modulation magnitude spectrum estimation, has shown promising results. (Paliwal et al., 2011) demonstrated that the MME approach, when compared with other approaches and with tuned parameters, showed the best performance in single-channel speech enhancement. The two other methods are the MMSE acoustic magnitude estimator of (Malah, 1984) and modulation spectral subtraction of (Paliwal et al., 2011).

(Godin, K.W. et al., 2013) conducted a comprehensive evaluation of multiple noise reduction and spectrum estimation techniques for noise-robust speaker identification. Their study revealed significant variations in the effectiveness of different methods. Notably, PSS, log-MMSE, MHEC, SWCE, and PMVDR resulted in significant relative improvements averaged across all tasks. In contrast, the Wiener filter and RLP led to significant degradation when averaged across all tasks. Interestingly, some techniques, including the G.729 window, LP-MFCCs, and FDLP, did not result in significant shifts when averaged across all tasks. These findings highlight the importance of careful selection of denoising techniques in speaker identification tasks.

In a separate study, (Sadjadi & Seyed, 2010) assessed single-channel speech enhancement techniques for speaker identification under mismatched conditions. They compared four distinct speech enhancement frontends: Spectral Subtraction (SS), Minimum Mean Square Error estimation (MMSE), Wiener filtering (WIN), and Principal Component Analysis in the Karhunen-Loève Transform domain (pKLT). Their results demonstrated that SS was the most successful among the four methods in mitigating the effect of mismatch on the performance of Speaker Identification (SID) systems. This was particularly evident at relatively low SNR levels (10 dB and below), suggesting that SS might be especially useful in challenging noise conditions.

2.2 Audio Feature

Feature extraction and selection play crucial roles in the development of effective models for speech emotion recognition (SER) and Automatic Personality Perception (APP), particularly in the context of Asynchronous Video Interviews (AVIs). This section of the literature review focuses on the various approaches to feature extraction and selection, as well as recent trends in the field.

2.2.1 Features Extraction

The audio descriptors can be broadly categorized into seven groups which are Intensity, Pitch, Loudness, Formants, Spectral features, Mel-Frequency Cepstral Coefficients (MFCCs) and Other features. Several studies have identified specific sets of features that are particularly useful for SER and personality prediction:

(Batrinca et al., 2011) used mean, min, max, and standard deviation of pitch and intensity in their study on automatic personality assessment.(Koutsoumpis et al., 2024) employed a comprehensive set of features including:13 Mel Frequency Cepstral Coefficients (MFCC 0 to 12), 12 Chromatograms (0–11), 12 Linear, Frequency Cepstral Coefficients (LFCC 0 to 11), 12 Mel Frequency Spectral Coefficients (MFSC 0 to 11), 7 Spectral contrasts (0–6), 5 Formants bandwidth (0–4), 5 Formants frequency (0–4), 5 Spectral parameters (bandwidth, centroid, flatness, flux, roll off), 2 Pitch (frequency, intensity), 2 Polynomial features (0, 1), Harmonic to noise ratio (HNR), Intensity, Root mean square (RMS), and Zero crossing rate.

(Mohammadi & Vinciarelli, 2012) focused on low-level prosodic features, including energy, pitch, first two formants, and length of voiced and unvoiced segments. Then four statistical properties are estimated:minimum, maximum, mean, and relative entropy of the differences between low-level feature values extracted from consecutive analysis windows. As there are six low-level features and four statistical properties, the resulting feature vector for a speech clip is 24 dimensions. (Gorbova et al., 2017) used MFCCs, zero crossing rate, speaking rate, and spectral energy distribution features (such as centroid, bandwidth, and contrast). These are commonly applied not only in emotion recognition field, but also in some researches related to personality analysis. In (Rangra et al., 2023), researchers used only the MFCC and the model achieved the accuracy from the classification results of 74% for emotions and 89% for Personality classifications. The classification results prove that MFCC can be considered as an effective feature for characterizing and recognizing emotions and personality simultaneously from audio.

Nisha et al. (2023) considered five features potentially associated with personality domains: response time (in milliseconds), intensity (in dB), pitch (in Hz), speech rate (words per minute), and discourse markers. These selections were made because these features are drawn from various domains of language use, reflecting interactional attributes (i.e.,response time), acoustic attributes (i.e., pitch and intensity), content or textual attributes(i.e., discourse markers), as well as paralinguistic (linguistic-external) speech attributes (i.e.,speech rate), examining them would allow us to consider language in its interaction with personality from a more holistic perspective. Results show that several speech features correspond to different personality dimensions.

Recently, end-to-end approaches involving deep neural networks have gained widespread popularity. (Palaz & Collobert, 2015) analyzed Convolutional Neural Networks (CNNs) for speech recognition, feeding temporal raw speech directly to the CNN instead of using pre-extracted feature vectors. They found that the first convolutional layer acts as a filter bank, automatically learning to extract relevant features from the unprocessed audio wave input. The key difference in this paper is that temporal raw speech is fed directly to the CNN instead of the conventional two step procedure of first extracting the feature vectors (e.g., MFCC) followed by feeding to a classifier. But in the noise robust ASR studies show that these features are susceptible to noise but not to the same extent as MFCC features (without any normalization). (Koutsoumpis et al., 2024) employed deeply learned embeddings that incorporated emotion information, capitalizing on transfer learning from emotion recognition tasks. They used the Wav2Vec CNN architecture, initially trained for speech recognition and then fine-tuned for emotion recognition using the Ravdess dataset. The authors state that they used these embeddings to "maximize the extracted information" beyond what handcrafted features could provide. These embeddings are expected to capture nuanced audio that might be relevant for personality prediction and emotion recognition. However, the document doesn't specify the exact dimensionality of these embeddings or how they are combined with the handcrafted features in the final model.

2.2.2 Feature selection

Feature selection is crucial for improving model performance and reducing computational complexity. (Polzehl & Metze, 2010) describe a feature selection process using Information Gain Ratio (IGR). They generated about 1450 features across the seven groups mentioned earlier. Features were ranked using Information Gain, which models the uncertainty of overall information and uncertainty reduction obtained by individual features. An optimal feature set size was determined by appending an increasing number of high-gain features into the feature space. Their analysis revealed the importance of MFCC-based features, along with features from intensity, duration of segments, and pitch derivatives.

2.3 Emotion Labeling

One of the most famous classification of emotions belongs to the American psychologist Robert Plutchik. In his taxonomy, there are 8 fundamental emotions (Joy, sadness, trust, disgust, fear, anger, anticipation, and surprise), grouped in 4 pairs of opposites. All of them manifest in various degrees of intensity and their combination results in secondary emotions. This complex scheme was graphically represented by Plutchik in a very suggestive chart, known ever since as ‘Plutchik’s wheel’ (LeDoux, 2001). The fundamental emotions in different degrees of intensity are associated with a large number of emotions and those represented in the central part of the wheel are considered to be the extreme intensity manifestations of these fundamental emotions. However, taking into consideration the complexity of human emotions, there is a general consensus regarding the fact that some of them are more important for communication and interaction with computers. Starting from Plutchik’s wheel, a simplified version (happiness, sadness, anger, fear, disgust, and surprise) is the reference in automatic emotion recognition (FRANT¸ et al., 2017).

In the context of job interviews, emotion labeling plays a crucial role in analyzing candidates' emotional states and responses. The basic emotions—happiness, sadness, anger, fear, disgust, and surprise—provide a foundation for understanding these emotional expressions. Research has shown that emotions like happiness are often linked to positive impressions in professional settings, as candidates who exhibit enthusiasm and optimism tend to be perceived more favorably (Cortez & Marshall, 2017). Conversely, sadness and anger can suggest dissatisfaction or emotional instability, which are generally viewed negatively in interviews (Tiedens, 2001) Moreover, fear is commonly associated with anxiety and nervousness, which may signal a lack of confidence, a critical factor in interview performance (McCarthy & Goffin, 2004). Disgust, while less likely to occur in interview settings, might convey distaste or dissatisfaction, further complicating the candidate's evaluation. Surprise, though generally neutral, could be interpreted as unpreparedness, especially in high-stakes scenarios like job interviews.

2.4 Personality Labeling

Personality assessment has been a significant area of research in psychology, with various theories developed to categorize and understand human personality. Over the years, several prominent personality theories have emerged, each offering unique perspectives on human personality. These include Cattell's Sixteen Personality Factor model (Cattell & Mead, 2008), Eysenck's Psychoticism, Extraversion, and Neuroticism model (Eysenck, 1981), and the Myers-Briggs Type Indicator (Soto & John, 2021). While these models have contributed significantly to our understanding of personality, research has shown limitations in their replicability and robustness (Digman, 1990). Cattell's 16-factor system, for instance, faced challenges in independent replication, with most studies finding fewer than 16 factors (Boyle, 1989). Similarly, Eysenck's three-factor approach, particularly the Psychoticism dimension, has been criticized as potentially a blend of other traits (ANGLEITNER et al., 1990). The Myers-Briggs Type Indicator (MBTI), despite its popularity, has faced scrutiny in academic research. Pittenger (1993) argues that the MBTI lacks sufficient empirical support and neglects fundamental psychological standards. The assertion of 16 distinct personality categories has been questioned, leading to concerns about its suitability for applications such as career planning.

In contrast, the Big Five model, also known as the Five-Factor Model (FFM), has emerged as the most widely accepted framework for personality assessment in contemporary research (XiaoMing, 2022). This model posits five robust, orthogonal traits that effectively capture personality attributes: Openness to Experience (the active seeking and appreciation of new experiences), Conscientiousness (the degree of organization, persistence, control, and motivation in goal-directed behavior), Extraversion (the quantity and intensity of energy directed outwards in the social world), Agreeableness (the kinds of interaction an individual prefers, ranging from compassion to tough-mindedness), and Neuroticism (an individual's propensity for psychological distress) (Favaretto et al., 2019). The Big Five model has gained prominence due to several key advantages. Unlike earlier models, it has demonstrated consistent replicability across various studies and cultures (Digman, 1990). It provides a broader, more nuanced framework for understanding personality, incorporating aspects of earlier theories into a more cohesive structure. Linguistic and interpersonal behavior research has validated the Big Five's stability and generalizability across different cultures. Moreover, the Big Five has proven effective across various domains, including automatic personality trait recognition (XiaoMing, 2022).

While several personality theories have contributed to our understanding of human personality, the Big Five model has emerged as the most reliable and comprehensive approach in contemporary personality research. Its replicability, robustness, and cross-cultural validity make it particularly suitable for applications in automatic personality trait recognition. As such, the Big Five model provides a solid theoretical foundation for research projects aiming to analyze and predict personality traits, especially in the context of digital interactions and automated assessments. This widespread acceptance and applicability of the Big Five model in both traditional psychological research and modern computational approaches to personality assessment underscore its value as a framework for understanding and predicting human personality traits.

2.5 Model Selection

The selection of appropriate machine learning algorithms is a critical factor in developing effective models for Speech Emotion Recognition (SER) and Automatic Personality Prediction (APP) in the context of Asynchronous Video Interviews (AVIs). Various studies have explored different algorithms and architectures, each with their own strengths and performance characteristics.

For SER, researchers have employed a range of approaches. The NPSO_CNN framework, as reported in (Rangra et al., 2023), achieved notable accuracies of 74% for emotion classification and 89% for personality classification. This demonstrates the potential of convolutional neural networks (CNNs) in capturing relevant features from speech data. Support Vector Machines (SVMs) have also shown promise in SER tasks. The study (Mohammadi & Vinciarelli, 2012) found that SVMs with Gaussian kernels provided improved accuracies compared to other traditional methods like Logistic Regression.

In the domain of APP, a variety of classifiers have been explored. The research (Gorbova et al., 2017) utilized Multilayer Perceptron Neural Networks, showcasing the applicability of deep learning approaches to personality prediction tasks. Another study of (Mohammadi et al., 2010), employed SVM with Radial Basis Function (RBF) kernel, which demonstrated balanced performance across different personality traits.

Comparative studies have also provided insights into algorithm selection. (Batrinca et al., 2011) compared Naïve Bayes, SVM with linear kernel, and SVM with RBF kernel. The results indicated that SVM-RBF was the only classifier yielding balanced performances across different traits, highlighting its potential for robust personality prediction.

The trend in recent SER and APP research points towards the use of sophisticated single-classifier architectures. These include advanced implementations of algorithms like SVM, particularly with RBF kernels, and various neural network architectures such as CNNs. These approaches have shown considerable promise in capturing the complex patterns necessary for accurate emotion recognition and personality prediction from speech data.

In conclusion, while algorithm selection heavily depends on the specific dataset and emotion or personality classes being considered, the literature indicates that SVMs (particularly with RBF kernels) and neural network-based algorithms like CNNs show the most potential for achieving high performance in both SER and APP tasks within the AVI context.

2.6 Chapter Summary and Evaluation

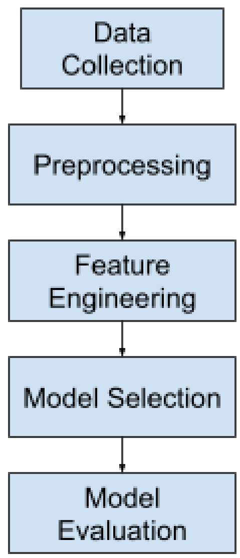
The content presented in this chapter provides a comprehensive overview of the key components involved in developing SER and APP models for AVIs. The Data Preprocessing section offers valuable insights into the importance of proper signal processing techniques, highlighting the need for careful consideration of framing and windowing options, as well as effective denoising methods. The discussion on various window sizes and step sizes across different studies shows the importance of experimentation to determine optimal configurations for specific applications.The Audio Feature section demonstrates the progress of feature extraction techniques, from traditional handcrafted features to more advanced deep learning approaches. The combination of both conventional acoustic features and newer embedding-based methods provides a balanced perspective on the current state of the field.The Emotion and Personality Labeling subsection successfully compares different emotion classification schemes and personality models. The focus on the Big Five model for personality assessment is well-justified, given its widespread acceptance and empirical support. The discussion on emotion labeling in the context of job interviews adds practical relevance to the theoretical frameworks presented.The Model Selection section provides a good overview of various algorithms used in SER and APP tasks, with a particular emphasis on SVMs and neural network-based approaches.

Chapter 3

Methodology and Requirements Analysis

3 Methodology and Requirements Analysis

This chapter outlines the methodological framework used in this study, detailing the processes and techniques employed to achieve the research objectives. It begins with an overview of the dataset and tools utilized, followed by a comprehensive explanation of the preprocessing steps necessary for preparing the data. Noise reduction techniques, including spectral subtraction, Wiener filtering, and MMSE denoising, are explained. The chapter also covers feature extraction methods, data structuring approaches, and the steps taken to ensure the reliability and validity of the processed data.



3.1 Data Collection

In this project, data collection involves three primary datasets: actual data from a company, a personality traits dataset, and an emotion dataset. Each dataset plays a unique role in training and evaluating models for Automatic Personality Perception (APP) and Speech Emotion Recognition (SER).

3.1.1 Actual data

The actual data consists of videos provided by the company, which contain question answer sessions from interviewers. This data serves as a sample for testing and allows us to evaluate the trained models on real world company data. By using these videos, the project can assess how well the models generalize beyond training datasets and can identify any potential adjustments needed to improve model performance. This real-world dataset ensures that the project outcomes are relevant and applicable to practical scenarios.

3.1.2 Personality Traits Dataset

The Chalearn First Impressions V2 dataset, sourced from ChaLearn's website, was utilized for data collection in this project. This dataset is specifically designed for research in Automatic Personality Perception (APP) and consists of 10,000 short video clips averaging 15 seconds each, extracted from over 3,000 YouTube videos. Each video features individuals facing the camera and speaking in English, representing diverse genders, ages, nationalities, and ethnicities. Each video clip is associated with six annotations (five traits and one job-interview variable), provided as a dictionary of dictionaries for structured and efficient data access.

The dataset is annotated with personality trait labels based on the Five-Factor Model (FFM), a widely accepted framework in personality research. The five personality traits Extraversion, Agreeableness, Conscientiousness, Neuroticism, and Openness are represented as continuous values within the range [0, 1]. These annotations were generated using a rigorous labeling process via Amazon Mechanical Turk (AMT) to ensure reliability. In addition to the personality trait labels, the dataset includes a "job-interview variable," indicating the likelihood of the person being invited to a job interview, also represented as a continuous value between [0, 1].

This dataset is critical for training the APP model, as it combines high-quality video data with reliable annotations for supervised learning. The diversity in demographic representation and the comprehensive labeling methodology enhance the dataset's applicability for building robust models that infer personality traits from visual and auditory cues. Audio data will be extracted using the moviepy library for further analysis and integration into the model.

3.1.3 Emotion Dataset

The emotion dataset used to train the Speech Emotion Recognition (SER) model is a combination of four well-known datasets which are CREMA-D, RAVDESS, SAVEE, and TESS. This combined dataset provides a diverse and comprehensive foundation for building an effective emotion classification model by capturing variations in speech across different demographics, emotions, and recording conditions.

- **CREMA-D (Crowd Sourced Emotional Multimodal Actors Dataset):** This dataset includes 7,442 clips from 91 actors (48 male, 43 female), aged 20 to 74, representing diverse races and ethnicities. Actors expressed six emotions (Anger, Disgust, Fear, Happy, Neutral, Sad) at varying levels of intensity (Low, Medium, High, Unspecified) across 12 different sentences.
- **RAVDESS (Ryerson Audio-Visual Database of Emotional Speech and Song):** This dataset contains 1,440 audio-only speech files performed by 24 professional actors (12 female, 12 male) speaking in a neutral North American accent. Emotions include Neutral,

Calm, Happy, Sad, Angry, Fearful, Disgust, and Surprise, with two intensity levels (Normal, Strong).

- **SAVEE (Surrey Audio-Visual Expressed Emotion):** SAVEE comprises recordings from four male speakers (aged 27–31) who expressed seven emotions (Anger, Disgust, Fear, Happiness, Sadness, Surprise, and Neutral). Each speaker recorded 15 sentences per emotion, resulting in a total of 120 utterances per speaker.
- **TESS (Toronto Emotion Speech Set):** This dataset includes 2,800 audio clips recorded by two actresses (aged 26 and 64) expressing seven emotions (Anger, Disgust, Fear, Happiness, Pleasant Surprise, Sadness, and Neutral). Each emotion was expressed through 200 target words embedded in the carrier phrase "Say the word _."

This combined dataset provides a diverse set of emotional expressions across genders, age groups, accents, and intensity levels, making it ideal for training a robust SER model. The integration of multiple datasets ensures variability in emotional representation while maintaining high-quality, controlled recordings. This diversity equips the model to accurately classify emotions from auditory cues and apply the insights to analyze emotional tones in real-world interview responses.

3.2 Preprocessing

Effective preprocessing is critical to ensure high-quality input data for training both the Automatic Personality Perception (APP) and Speech Emotion Recognition (SER) models. This process prepares the data by improving audio clarity, standardizing formats, and reducing noise. Key preprocessing techniques include resampling, noise reduction, and normalization, along with evaluations to measure the effectiveness of these methods.

3.2.1 Resampling

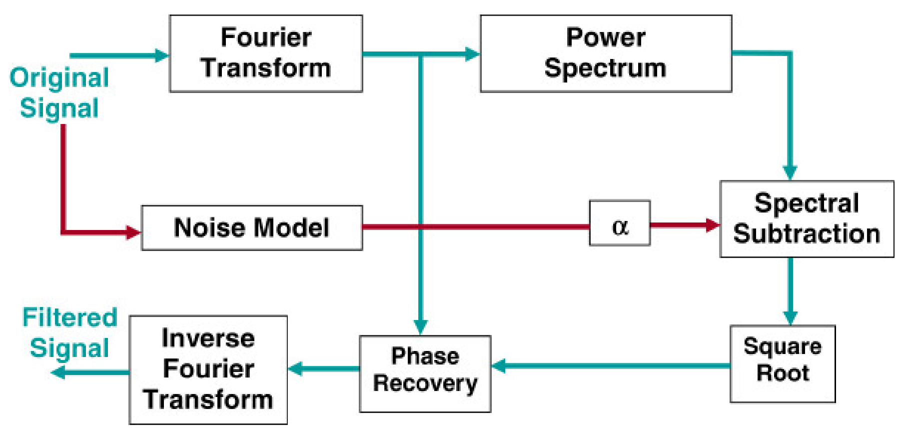
In this project, all audio data is resampled to a sampling rate of 16,000 Hz. This sampling rate is chosen because it is widely used in speech processing and provides a balance between computational efficiency and audio quality. A 16 kHz rate effectively captures the relevant frequency range for human speech while reducing the file size compared to higher rates such as 44.1 kHz, which may include unnecessary high frequencies that are less relevant to speech analysis.

3.2.2 Noise Reduction

Noise reduction is a critical preprocessing step in audio analysis that involves minimizing or eliminating unwanted noise from signals to enhance their clarity and quality. This step is essential in applications such as speech recognition, emotion detection, and audio classification, where background noise can degrade the performance of subsequent analysis. Noise reduction algorithms operate by analyzing the noise characteristics and suppressing them while preserving the original signal as much as possible.

1. **Spectral Subtraction**

   This method assumes that noise remains stationary or varies slowly compared to the speech signal.



- The **Short-Time Fourier Transform (STFT)** is applied to the noisy signal to obtain its magnitude and phase.

- The **noise spectrum** is estimated by averaging the magnitude spectrum of initial frames containing only noise.
- The **clean signal spectrum** is computed by subtracting a scaled noise spectrum (α) from the noisy spectrum.
- The clean spectrum is combined with the original phase, and the signal is reconstructed using the **inverse STFT (ISTFT)**.

2. **MMSE (Minimum Mean Square Error) Denoise**
   This technique estimates clean signal power using a probabilistic approach.
   - The STFT of the noisy signal is computed, and the power spectral density (PSD) of the noise is estimated from noise-only frames.
   - The MMSE gain function suppresses noise while minimizing speech distortion:

$$G(f) = \max\left(1 - \frac{\mathrm{PSD}_{\mathrm{noise}}}{\mathrm{PSD}_{\mathrm{speech}} + \epsilon}, 0\right)$$

   where ϵ is a small constant to prevent division by zero.

   - The clean signal is reconstructed by applying the gain to the noisy signal in the frequency domain.

3. **Wiener Filter**
   Based on the principle of minimizing the mean square error between the clean and estimated signals, Wiener filtering is another effective method:

   - The spectrogram of the noisy signal is computed to separate its magnitude and phase components.
   - The **noise power** is estimated, and the **signal power** is derived by subtracting the noise power from the total power:

$$P_{\mathrm{signal}} = \max(P_{\mathrm{noisy}} - P_{\mathrm{noise}}, 0)$$

   - A Wiener filter is applied:

$$H(f) = \frac{P_{\text{signal}}}{P_{\text{signal}} + P_{\text{noise}}}$$

- The filtered magnitude spectrum is combined with the original phase to reconstruct the clean signal using ISTFT.

4. **NoiseReduce Library**
The **NoiseReduce** library in Python offers a simplified interface for noise reduction, providing multiple noise reduction techniques including spectral gating. This library estimates noise during silent sections and then reduces it adaptively during speech. It is flexible and efficient for rapid deployment in projects where simplicity and adaptability are needed.

To evaluate the effectiveness of noise reduction, we use several objective metrics that measure audio quality and intelligibility:

1. **PESQ (Perceptual Evaluation of Speech Quality)**
PESQ is an objective measure that predicts subjective quality scores. It compares the processed signal to a reference signal and computes a quality score ranging from -0.5 to 4.5.
2. **STOI (Short-Time Objective Intelligibility)**
STOI is a measure of speech intelligibility and focuses on how understandable the speech is after processing. It calculates a correlation coefficient between the original and processed audio spectrograms, with values ranging from 0 (poor intelligibility) to 1 (high intelligibility).
3. **SegSNR (Segmental Signal-to-Noise Ratio)**
SegSNR computes the average SNR over short time segments, providing a measure of how well the noise has been suppressed in each segment. Higher SegSNR values indicate better noise reduction. This metric helps identify if the noise reduction is consistent across the audio signal.
4. **LSD (Log-Spectral Distance)**
LSD measures the log-spectral deviation between the clean and processed signals, offering insight into how much the spectral characteristics have changed. A lower LSD indicates less spectral distortion, implying that the noise reduction technique has preserved the original signal well.

3.2.3 Normalisation

Normalization adjusts the amplitude of audio signals to a standard range, ensuring uniformity across datasets. This step is crucial in preparing the data for feature extraction, as it removes amplitude variability that could bias the model. Typically, normalization involves scaling each audio signal to have zero mean and unit variance or using peak normalization to set the maximum amplitude to a specific value, such as 1. Normalization enhances the robustness of the model by reducing variations due to recording conditions, speaker volume, and other external factors, allowing the model to focus on the core features of the audio.

3.3 Feature Engineering

3.3.1 Feature Extraction

The feature extraction process for audio analysis is divided into four main categories: prosodic features, spectral features, voice quality features, and statistical summaries. There are a total 171 hand-crafted features extracted. The extraction has been performed with Praat and Librosa. Four statistical properties are estimated:mean, minimum, maximum, and standard deviation between low-level feature values extracted from the audio. Table below shows the features extracted from the audio signal.

Prosodic Features	Description
Intensity	measure of the signal's loudness
Pitch	represents the perceived frequency of the sound
Speech rate	number of zero-crossings per second, the speed at which a speaker is talking
Energy	speech signal's power or loudness over time

Table 3.3.1.1 Prosodic Feature extracted

Spectral Feature	Description
Mel-frequency cepstral coefficients (MFCC)	capture the short-term power spectrum of the speech
Chroma	represents the energy content in each of the 12 pitch classes, computed from the spectrogram
Spectral Contrast	measures the difference in amplitude between peaks and valleys in a sound’s spectrum and reflects the timbral texture
Spectral Centroid	capture the center of the spectrum
Spectral bandwidth	capture the spread of the spectrum
Spectral flatness	indicates how noise-like or tonal a sound is
Spectral rolloff	Frequency below which a certain percentage of total spectral energy lies.

Linear Predictive Coding (LPC)	Estimate the vocal tract shape

Table 3.3.1.2 Spectral Feature extracted

Voice Quality Feature	Description
Formants	represent resonant frequencies in the human vocal tract
Harmonic-to-Noise Ratio (HNR)	ratio of harmonics to noise in the voice, reflecting the breathiness or hoarseness of the speaker's voice
Root mean square energy (RMS)	measure of the magnitude of a signal
Zero Crossing Rate	Number of times the signal changes sign.

Table 3.3.1.3 Voice Quality Feature extracted

Statistical Summaries	Description
Mean	The average value of the feature across the frame
Minimum	The minimum value of the feature across the frame.
Maximum	The maximum value of the feature across the frame.
Standard deviation	The standard deviation of the feature across the frame

Table 3.3.1.4 Statistical Summaries

3.3.2 Feature Selection

To select the most relevant features for personality and emotion prediction from the extracted dataset, an entropy-based feature ranking approach is utilized. This involves assessing the information gain (IG) of each feature with respect to the target classes. Information Gain (IG) measures the reduction in uncertainty or entropy achieved by knowing the value of a feature.

$$IG(Y,X) = E(Y) - E\left(\frac{Y}{X}\right)$$

E : Entropy

Y : *dependent variable*

X : *independent variable*

Using this metric, features are ranked based on their ability to identify between personality traits and emotions. The feature space is expanded incrementally, appending the highest ranked features based on IG until the classification performance peaks.

3.4 Data Preparation

3.4.1 Sampling

upsampling is applied to handle class imbalance. The dataset is separated into individual DataFrames based on unique labels, and the class with the maximum size is identified. Then, each class is upsampled using the resample function from sklearn.utils, where the smaller classes are randomly sampled with replacement to match the size of the largest class. This step ensures that each class has an equal number of samples, reducing bias towards the larger classes and allowing the model to learn from all classes equally. After upsampling, the individual DataFrames are combined back into a single DataFrame, and the rows are shuffled to randomize the data order. The final upsampled dataset is reset, ensuring no residual index issues.

3.4.2 Encode

label encoding is applied to convert categorical labels into numeric values using LabelEncoder from sklearn.preprocessing. This step is essential for transforming non-numeric labels, such as strings, into a format that can be processed by machine learning algorithms. The encoded labels are then stored in a variable to be used in model training. In the context of the Big Five personality traits, label encoding is used to convert the categorical personality labels (such as "Extraversion," "Agreeableness," "Conscientiousness," "Neuroticism," and "Openness") into numeric values that can be processed by machine learning algorithms. For instance, each of the five traits would be assigned a unique numeric value to represent them in the model's dataset.

3.4.3 Scaling

feature scaling is performed to ensure that all input features are on a comparable scale. This is achieved using the MinMaxScaler, which scales the data into the range [0, 1], making it suitable for models that are sensitive to the magnitude of the features. By scaling the features, the model can learn more effectively, avoiding any one feature from dominating the learning process due to its larger scale.

3.4.4 Train-test split

The dataset is split into training and testing sets using train_test_split from sklearn.model_selection. The training set is used to train the model, while the test set is kept aside to evaluate the model's performance on unseen data. Typically, a portion, such as 80% of the data, is allocated for training, and the remaining 20% is used for testing. The split is randomized, with the random state set to ensure that the process is reproducible. The result is a well-prepared dataset where labels are balanced, features are scaled, and the data is ready for training and testing.

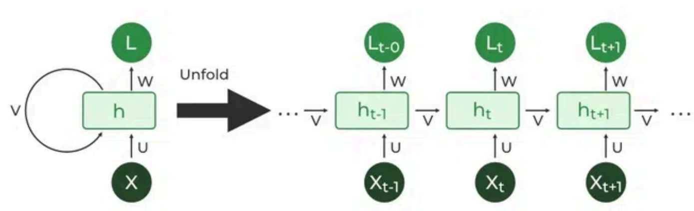
3.5 Model Selection

In model selection, Recurrent Neural Networks (RNNs), Extreme Gradient Boosting (XGBoost), Random Forest and Support Vector Machine (SVM) will be used to train on the feature set and evaluate their performance. Ensemble model which combines these 4 models will be used in the final development for predicting the company's actual dataset to reduce the generalization error of the prediction..

3.5.1 Recurrent Neural Networks (RNNs)

Recurrent neural networks are a class of artificial neural networks introduced in the 1980s by researchers David Rumelhart, Geoffrey Hinton, and Ronald J. Williams. Unlike feedforward neural networks, which process data in a single pass, RNNs process data across multiple time steps, making them well-adapted for modeling and processing text, speech, and time series.

RNNs are designed to handle sequential data by maintaining a hidden state that captures information about previous inputs. This makes them highly effective for analyzing temporal patterns, such as those present in speech signals. The architecture of an RNN involves repeating layers where each time step’s output depends on both the current input and the previous hidden state.



The hidden state update and output are expressed as:

$$h = \sigma(U \cdot X + W \cdot h_{t-1} + B)$$

Here, h represents the current hidden state, U and W are weight matrices, and B is the bias.

$$Y = O(V \cdot h + C)$$

The output Y is calculated by applying O, an activation function, to the weighted hidden state, where V and C represent weights and bias.

3.5.2 Gradient Boosting Models (GBMs)

Extreme Gradient Boosting or XGBoost, is a high-performance implementation of gradient boosting that is optimized for speed and accuracy. It enhances traditional Gradient Boosting Machines (GBMs) by incorporating advanced techniques like regularization, parallel processing, and tree pruning. These improvements make it highly effective for structured data tasks like feature selection and classification.

XGBoost operates on the principle of boosting by sequentially adding decision trees to correct the errors of the previous iterations. The process begins with initialization, where a base prediction is set, such as the mean of the target variable for regression tasks or the log odds for classification problems. For each boosting round, the algorithm calculates the gradient and the Hessian, which are the first and second derivatives of the loss function, respectively. These values quantify how the model's predictions should be adjusted to minimize the objective function effectively.

The core of XGBoost lies in constructing decision trees using these gradients and Hessians. At each tree node, the algorithm determines the optimal split by maximizing a metric called **gain**, which measures the improvement in the objective function due to the split. The formula for gain is:

$$Gain = \frac{1}{2}\left[\frac{G_L^2}{H_L+\lambda} + \frac{G_R^2}{H_R+\lambda} - \frac{(G_L+G_R)^2}{H_L+H_R+\lambda}\right] - \gamma$$

Here, GL, GR, GT are the gradients for the left, right, and total nodes, respectively, while HL, HR, HT are the corresponding Hessians. The parameters λ and γ add regularization to prevent overfitting by discouraging overly complex trees.

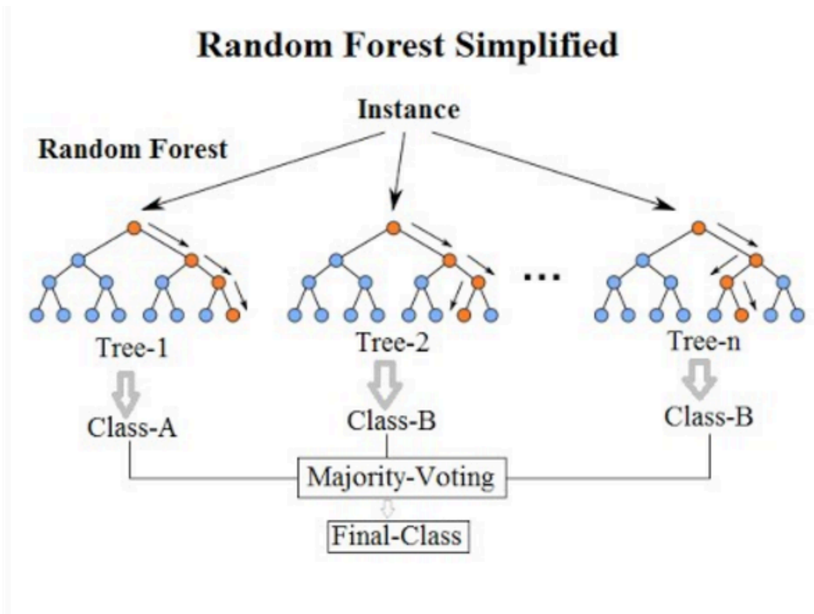
Once a tree is constructed, its predictions are scaled by a learning rate (η) and added to the model's cumulative predictions:

$$\begin{aligned}
\hat{y}_i^{(0)} &= 0 \\
\hat{y}_i^{(1)} &= f_1(x_i) = \hat{y}_i^{(0)} + f_1(x_i) \\
\hat{y}_i^{(2)} &= f_1(x_i) + f_2(x_i) = \hat{y}_i^{(1)} + f_2(x_i) \\
&\cdots\cdots \\
\hat{y}_i^{(t)} &= \sum_{k=1}^{t} f_k(x_i) = \hat{y}_i^{(t-1)} + f_t(x_i)
\end{aligned}$$

This process of updating predictions continues, with additional trees added iteratively until a specified stopping criterion is met, such as reaching the maximum number of iterations or observing no further improvement in the validation set. By combining the strengths of gradient-based optimization and regularization, XGBoost achieves high accuracy and robustness in predictive modeling.

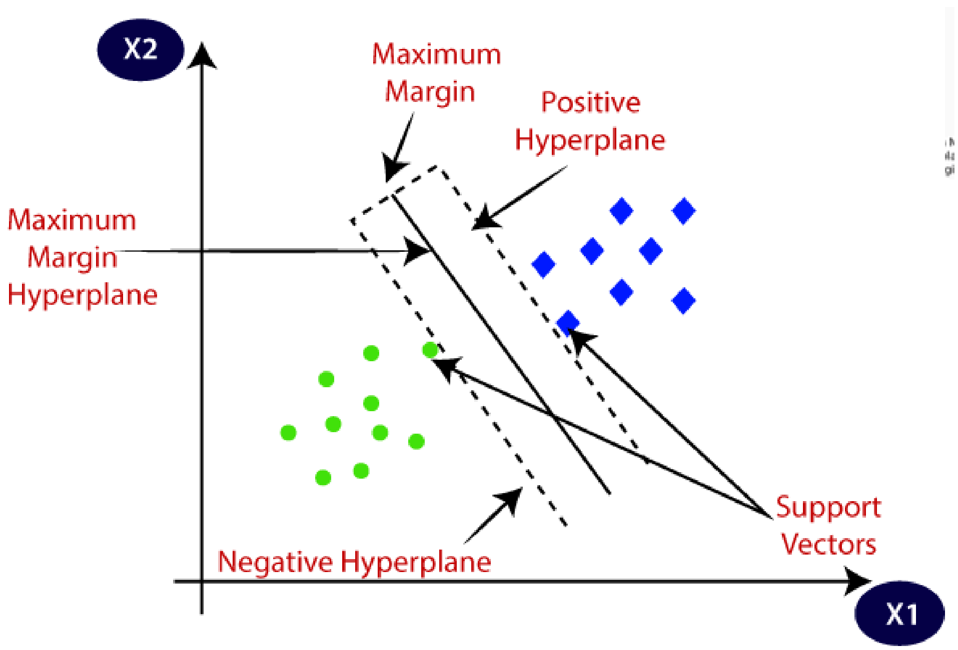
3.5.3 Random Forest

Random forest is a popular machine learning algorithm developed by Leo Breiman and Adele Cutler. It is suitable for both classification and regression tasks. It operates by building multiple decision trees during training and combining their outputs to enhance the predictive performance and control overfitting. Each tree is constructed using a random subset of the data, and at each node, a random subset of features is selected for splitting, introducing diversity among the trees. The final prediction is made by taking the majority vote in classification tasks. Random Forest is robust to noisy data and capable of handling large datasets with high dimensionality. Additionally, it provides feature importance scores, aiding in understanding the contribution of each feature to the model.



3.5.4 Support Vector Machine (SVM)

Support Vector Machine (SVM) is a supervised learning algorithm effective for both linear and nonlinear classification tasks. SVM works by finding the optimal hyperplane that separates data points of different classes in the feature space. For linearly inseparable data, it employs a kernel trick, such as radial basis function (RBF) kernels, to project the data into a higher-dimensional space where a linear hyperplane can be found. The goal of SVM is to maximize the margin between the nearest data points of each class, known as support vectors, ensuring robustness to small perturbations. SVM performs well in high-dimensional spaces and is memory efficient as it uses a subset of training points (support vectors) for decision-making.



3.6 Model Evaluation

To assess model performance effectively, a range of metrics is used to capture different aspects of predictive accuracy, error, and reliability. These metrics provide insights into how well the model generalizes to new data, its ability to distinguish between classes, and the overall error associated with its predictions. This section details the key metrics used: accuracy, precision, recall, F1-score, mean absolute error (MAE), mean squared error (MSE), and root mean squared error (RMSE).

3.6.1 Accuracy

Accuracy is the ratio of correctly predicted instances to the total instances. Accuracy is useful when the dataset is balanced; however, it can be misleading in imbalanced datasets, as it may reflect a high score even if the model mostly predicts the majority class.

$$Accuracy = \frac{TN + TP}{TN + FP + TP + FN}$$

3.6.2 Precision

Precision measures the proportion of true positive predictions among all positive predictions. It indicates the accuracy of the model's positive predictions and is critical in scenarios where false positives are costly. A high precision score indicates that the model has a low rate of false positives.

$$Precision = \frac{TP}{TP + FP}$$

3.6.3 Recall

Recall, or sensitivity, measures the proportion of actual positive instances correctly identified by the model. This metric is vital when it is important to minimize false negatives. A high recall score indicates that the model captures most of the positive instances, though it may come at the cost of some false positives.

$$Recall = \frac{TP}{TP + FN}$$

3.6.4 F1-Score

The **F1-Score** is the harmonic mean of precision and recall, balancing the trade-off between these two metrics. It provides a single score to assess the model's performance when both precision and recall are important. The F1-score is particularly valuable for imbalanced datasets where focusing solely on accuracy might be misleading.

$$F1\ Score = 2 * \frac{Precision * Recall}{Precision + Recall}$$

3.7 Chapter Summary and Evaluation

At the conclusion of this chapter, the methodologies discussed are evaluated in relation to their effectiveness and relevance to the research objectives. The data preprocessing steps, particularly noise reduction techniques like spectral subtraction, Wiener filtering, and MMSE denoising, are critically analyzed for their ability to enhance signal clarity and preserve essential features. The feature extraction and data structuring methods are also evaluated for their alignment with the goals of the analysis, ensuring the processed data retains its representational accuracy. Additionally, the integration of these methodologies into a cohesive framework is reviewed to confirm the logical progression from data preparation to analytical implementation.

Chapter 4

Experiment and result

4 Experiment and Result

```
Label distribution:
label
openness             3536
agreeableness        2651
conscientiousness    2140
neuroticism          1132
extraversion          541
Name: count, dtype: int64
```

Figure 4.0.1 original personality label distribution

```
New label distribution after upsampling:
label
agreeableness        3536
extraversion         3536
openness             3536
neuroticism          3536
conscientiousness    3536
Name: count, dtype: int64
```

Figure 4.0.2 personality label distribution after upsampling

For the personality dataset, the experiment began by downloading the video dataset from the official ChaLearn website, which contains 6,000 training samples, 2,000 test samples, and 2,000 validation samples. All video data were first combined, and the corresponding audio data were extracted for feature analysis. The ground truth file provided by ChaLearn was used to annotate the labels for each sample. Initially, each data point had personality traits represented by multiple numerical values, the trait with the highest value was selected as the label for that sample when preprocessing. Following this, data preprocessing steps were applied to prepare the data for model training. Figure 4.0.1 showed the original label distribution, which was highly imbalanced. To address this, upsampling techniques were employed to balance the dataset, as shown in Figure 4.0.2. After balancing, the data underwent encoding and scaling processes to ensure it was suitable for model training. Finally, the preprocessed data was split into an 80-20 ratio for training and testing purposes. This structured preparation ensures that the data is ready for robust model development and evaluation.

```
Emotion
happy       1923
sad         1923
angry       1923
fear        1923
disgust     1923
neutral     1703
surprise     652
calm         192
Name: count, dtype: int64
```

Figure 4.0.3 original emotion label distribution

```
Updated label distribution after upsampling:
Emotion
neutral    1923
happy      1923
sad        1923
angry      1923
fear       1923
disgust    1923
Name: count, dtype: int64
```

Figure 4.0.4 emotion label distribution after preprocess

For the emotional dataset, the audio data with corresponding labels was already prepared and made available on Kaggle. The dataset was downloaded, and the audio data underwent necessary preprocessing as mentioned in chapter 3. Since the labels were already provided, no additional annotation was required. During data preparation, the labels "calm" and "surprise" were removed due to their low representation in the dataset, which could negatively impact the model's training and evaluation. To address the class imbalance among the remaining labels, upsampling techniques were applied to ensure that all classes were balanced as shown in figure 4.0.4. Once preprocessing was completed, the data was split into an 80-20 training and testing set to train and evaluate the models effectively.

After training the model, the performance was evaluated using appropriate metrics such as accuracy, precision, recall, F1-score, and other relevant measures, depending on the task. The model was trained on the extracted features, including prosodic features (pitch, intensity, energy, and speech rate), spectral features (MFCC, spectral centroid, bandwidth, roll-off, and chroma), and voice quality features (formants, Harmonic-to-Noise Ratio, and zero-crossing rate), as described earlier. These features served as the input to the model, enabling it to capture both high-level and low-level speech characteristics.
For further clarity, a confusion matrix was generated to analyze the model's misclassifications, highlighting specific areas where the model performed well and where further improvements may be necessary. Additionally, ROC curves and AUC scores were included to provide insights into the model's ability to discriminate between classes. Visualizations such as accuracy and loss

curves during training were also plotted, showing the model's learning process and convergence over epochs. This helped in understanding whether the model overfit, underfit, or generalized well to unseen data.

In summary, the results demonstrated the potential of the trained model in analyzing speech features for tasks such as emotion recognition or personality detection. Limitations, such as class imbalance or misclassifications, were also discussed, with recommendations for future work to enhance performance.

4.1 Exploring Speech Emotion Recognition model

4.1.1 RNN Model

	precision	recall	f1-score	support
angry	0.73	0.75	0.74	391
disgust	0.66	0.51	0.57	373
fear	0.68	0.54	0.60	398
happy	0.71	0.51	0.60	379
neutral	0.51	0.77	0.61	398
sad	0.59	0.69	0.63	369
accuracy			0.63	2308
macro avg	0.65	0.63	0.63	2308
weighted avg	0.65	0.63	0.63	2308

Figure 4.2.1.1 Classification Report of RNN

```
Average Accuracy: 0.6065
Average Precision: 0.6160
Average Recall: 0.6065
Average F1 Score: 0.6043
```

Figure 4.2.1.2 KFold result of RNN

The RNN model achieved an Accuracy of 62.91% on the test set, with a weighted Precision of 64.69%, a weighted Recall of 62.91%, and a weighted F1 Score of 62.73%. Using k-fold cross-validation, the average accuracy improved slightly to 65.05%, while the average precision, recall, and F1 scores were 65.11%, 65.05%, and 64.86%, respectively. This indicates that while RNN demonstrates moderate performance, it shows variability across folds, suggesting scope for further optimization.

The RNN model performed best for the angry class, achieving the highest precision of 73% and recall 75%, with an F1-score of 74%. For the neutral class, the model achieved a high recall of 77% but at the cost of lower precision 51%, indicating that it predicted neutral emotions more frequently, even when the actual label differed. The disgust, happy and fear classes had lower recall scores of 51% and 54% respectively, showing the model's difficulty in identifying these emotions accurately.

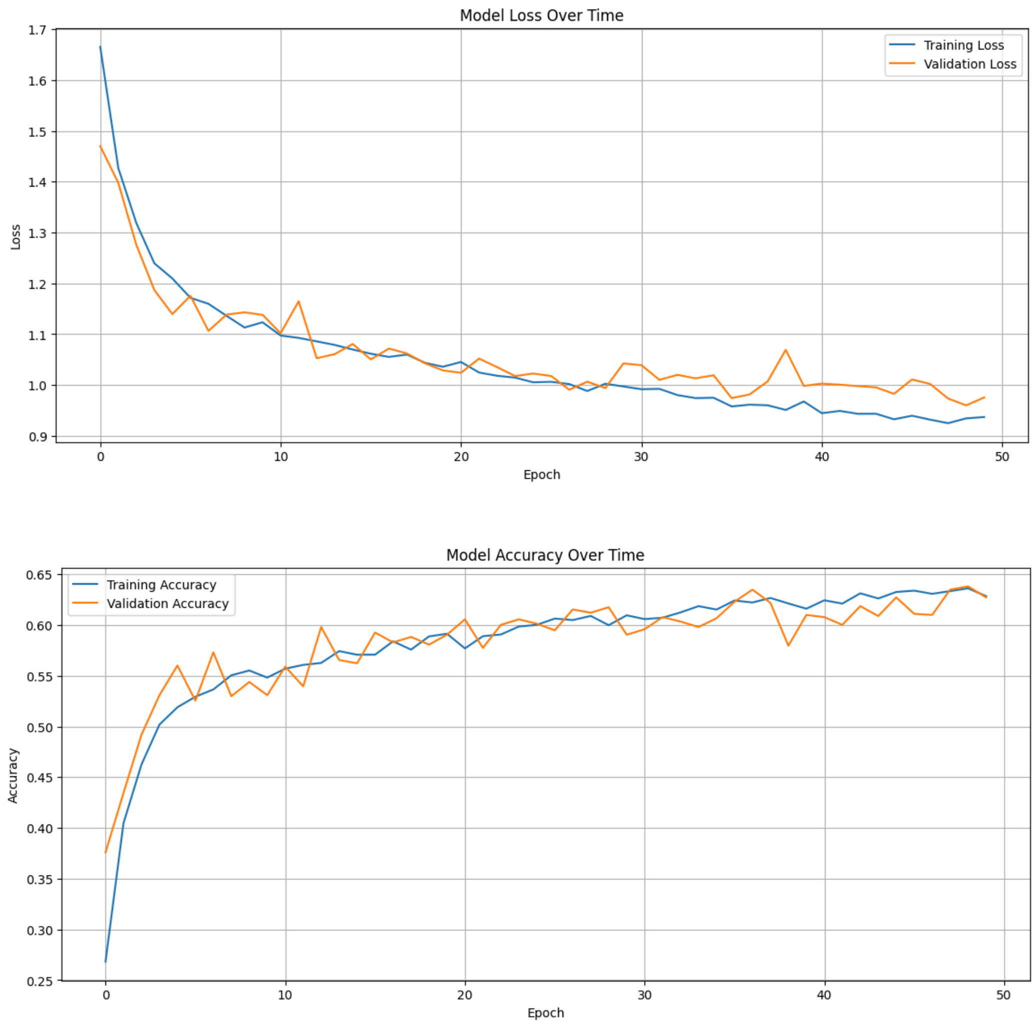


Figure 4.1.1.2 loss and accuracy curve Of RNN

In figure 4.1.1.2, it shows that both the training and validation loss decreases sharply during the initial episodes, indicating that the model is learning effectively. Around epoch 15, the losses start to stabilize, with the training loss decreasing at a slower rate. In addition, the validation loss exhibits minor fluctuations, suggesting slight overfitting towards the later epochs. In the final phase, the training loss reaches approximately 0.92, and the validation loss stabilizes around 0.96. The close proximity between training and validation losses indicates that the model generalizes reasonably well but shows some overfitting.

4.1.2 XGBoost

	precision	recall	f1-score	support
angry	0.82	0.90	0.86	391
disgust	0.78	0.65	0.71	373
fear	0.88	0.67	0.76	398
happy	0.82	0.80	0.81	379
neutral	0.72	0.83	0.78	398
sad	0.74	0.87	0.80	369
accuracy			0.79	2308
macro avg	0.79	0.79	0.79	2308
weighted avg	0.79	0.79	0.79	2308

Figure 4.2.2.1 classification report of XGBoost

```
Average Accuracy: 0.5968
Average Precision: 0.5992
Average Recall: 0.5968
Average F1 Score: 0.5892
```

Figure 4.2.2.2 KFold result of XGBoost

XGBoost achieved an Accuracy of 78.77%, with a weighted Precision of 79.36%, a weighted Recall of 78.77%, and a weighted F1 Score of 78.55%. However, k-fold cross-validation results were significantly lower, with an average accuracy of 66.00%, average precision of 65.86%, average recall of 66.00%, and average F1 score of 65.62%. This discrepancy suggests that the model is overfitting to the test set and requires further evaluation on the unseen data.

The angry class demonstrates strong performance, particularly in recall of 90%, which indicates that most instances of anger were correctly identified. However, the slightly lower precision 82% indicates that some non angry instances were incorrectly classified as angry. The disgust class shows a moderate level of performance with relatively low recall of 65%. This suggests the model struggles to identify instances of disgust accurately, with many true disgust instances being misclassified as another emotion. The fear class shows high precision of 88%, indicating that most of the predictions labeled as fear are accurate. However, recall of 67% suggesting that many fear instances are misclassified into other categories. The happy class has a balanced performance across precision and recall, suggesting the model handles this class effectively. The F1-score of 0.81 reflects strong overall performance in identifying happy emotions.The neutral class shows higher recall 83% than precision 72%, indicating that most neutral instances are correctly classified. However, the lower precision suggests that other emotions, particularly those with overlapping features like sadness or disgust, may sometimes be misclassified as neutral.The sad class exhibits strong recall, meaning the model identifies most instances of sadness accurately. The lower precision of 74% indicates that a noticeable proportion of instances from other emotions, particularly fear or neutral, may be misclassified as sad.

4.1.3 Random Forest

	precision	recall	f1-score	support
angry	0.76	0.91	0.83	391
disgust	0.88	0.73	0.80	373
fear	0.97	0.73	0.83	398
happy	0.83	0.80	0.81	379
neutral	0.84	0.94	0.88	398
sad	0.79	0.90	0.84	369
accuracy			0.83	2308
macro avg	0.84	0.83	0.83	2308
weighted avg	0.84	0.83	0.83	2308

Figure 4.2.3.1 classification report of Random Forest

```
Average Accuracy: 0.5994
Average Precision: 0.6064
Average Recall: 0.5994
Average F1 Score: 0.5904
```

Figure 4.2.3.2 KFold result of Random Forest

The Random Forest model achieved a respectable Accuracy of 83.41% on the test set, with a weighted Precision of 84.47%, a weighted Recall of 83.41%, and a weighted F1 Score of 83.29%. However, the k-fold cross-validation results revealed lower performance, with an average accuracy of 60.60% and average F1 score of 59.60%. This drop in performance highlights potential inconsistencies across the dataset splits, suggesting that the model is overfitting and the model's generalization ability may be limited.

The model demonstrates high recall for the angry class, correctly identifying 91% of true angry instances. However, the lower precision of 76% suggests that a relatively higher number of non angry instances are being misclassified as angry. The disgust class shows strong precision, indicating that most of the predictions labeled as disgust are accurate. However, 27% of true disgust instances are being misclassified as other emotions.The fear class exhibits excellent precision of 97%, meaning almost all predictions of fear are correct. However, the relatively low recall of 73% suggests that many true fear instances are being misclassified, potentially due to confusion with sad or disgust. The happy class maintains balanced performance across precision and recall. This indicates that the Random Forest model handles positive emotions effectively and has no major issues in this category. The neutral class shows strong performance, particularly in recall of 94%, indicating that almost all neutral instances are correctly classified. Precision is slightly lower at 84%, suggesting that some other emotions, such as sad, may occasionally be misclassified as neutral.The sad class exhibits strong recall , meaning most instances of sadness are correctly identified. The slightly lower precision 79% indicates that there are occasional misclassifications of other emotions, such as fear or neutral, as sad.

4.1.4 SVM

	precision	recall	f1-score	support
angry	0.83	0.88	0.86	391
disgust	0.74	0.70	0.72	373
fear	0.78	0.69	0.73	398
happy	0.81	0.77	0.79	379
neutral	0.78	0.81	0.79	398
sad	0.73	0.82	0.77	369
accuracy			0.78	2308
macro avg	0.78	0.78	0.78	2308
weighted avg	0.78	0.78	0.78	2308

Figure 4.2.4.1 classification report of SVM

```
Average Accuracy: 0.6419
Average Precision: 0.6408
Average Recall: 0.6419
Average F1 Score: 0.6404
```

Figure 4.2.4.2 KFold result of SVM

The SVM model achieved a test set Accuracy of 77.86%, with a weighted Precision of 77.91%, a weighted Recall of 77.86%, and a weighted F1 Score of 77.77%. The k-fold cross-validation results showed an average accuracy of 65.35%, with precision, recall, and F1 scores of 65.28%, 65.35%, and 65.22%, respectively. This suggests that SVM provides relatively balanced performance and maintains consistency across folds.

The model performs well for the angry class, with high precision and recall. This indicates it accurately predicts angry emotions from others. The disgust class precision and recall are relatively low, indicating difficulty in correctly predicting disgust and distinguishing it from related emotions like fear or angry.The fear class shows modest precision but suffers from a lower recall, meaning that a significant proportion of true fear instances are misclassified as other emotions. The F1-score of 0.73 reflects this trade off between precision and recall.The performance for happy is balanced and relatively strong. SVM captures this positive emotion fairly well but occasionally misclassifies it, likely as neutral or sad. The neutral class shows solid performance, with high recall indicating that most neutral instances are correctly classified. Precision is slightly lower, suggesting occasional misclassification of other emotions as neutral. For the sad class, SVM exhibits a high recall of 82%, capturing most true sad instances. However, its lower precision at 73% indicates that other emotions may sometimes be misclassified as sad.

4.1.5 Overall

	Accuracy	Precision	Recall	F1-Score
RNN	62.91	64.69	62.91	62.73
K-Fold RNN	60.65	61.60	60.65	60.43
XGBoost	78.77	79.36	78.77	78.55
K-Fold XGBoost	59.68	59.92	59.68	58.92
Random Forest	83.41	84.47	83.41	83.29
K-Fold RF	59.94	60.64	59.94	59.04
SVM	77.86	77.91	77.86	77.77
K-Fold SVM	64.19	64.08	64.19	64.04

The results of the emotion recognition models reveal a range of performances across different algorithms, with Random Forest demonstrating the highest test accuracy (83.41%), followed by XGBoost (78.77%), SVM (77.86%), and RNN (62.91%). However, k-fold cross-validation results indicate a general trend of overfitting for all of the models that exhibit more consistent but moderate performances.

Each model has strengths and weaknesses for different emotion classes. For instance, Random Forest performs best overall but struggles with precision for certain emotions. XGBoost achieves high recall for the angry class but misclassifies some emotions, while SVM demonstrates balanced performance but lower recall for fear and disgust. The RNN model, while showing potential with high recall for neutral and angry classes, struggles significantly with emotions like disgust and fear.

While each model exhibits strengths, further optimization is needed to ensure robust and generalizable performance. By incorporating the techniques, particularly focusing on feature engineering, hyperparameter tuning, and class balancing, model performance can be significantly improved.

4.2 Exploring Automatic Personality Perception model

4.2.1 RNN

	precision	recall	f1-score	support
agreeableness	0.66	0.62	0.64	708
conscientiousness	0.62	0.77	0.69	713
extraversion	0.91	0.98	0.94	726
neuroticism	0.81	0.84	0.82	697
openness	0.70	0.47	0.56	692
accuracy			0.74	3536
macro avg	0.74	0.74	0.73	3536
weighted avg	0.74	0.74	0.73	3536

Figure 4.2.1.1 Classification Report of RNN

```
Average Accuracy: 0.7137
Average Precision: 0.7074
Average Recall: 0.7137
Average F1 Score: 0.7059
```

Figure 4.2.1.2 KFold result of RNN

The RNN model demonstrates decent performance on the test set with an accuracy of 74.21% t, with a weighted Precision of 73.51%, a weighted Recall of 74.21%, and a weighted F1 Score of 73.51%. The average accuracy from k-fold cross-validation drops to 71.37%, about a 2.84% decrease compared to the test accuracy. This drop indicates mild overfitting, as the model performs slightly better on the test data compared to its generalization on unseen folds.

Extraversion achieved the highest performance with an F1 score of 94%. This outstanding performance can be attributed to the upsampling technique applied during preprocessing, which duplicated the samples of this label to ensure class balance. As a result, the model was exposed to more instances of the extraversion label during training, allowing it to better identify and predict this personality trait. Neuroticism also performed well, with an F1 score of 82%. This indicates that the model is able to distinguish this class effectively. Conscientiousness showed a high recall of 77% but a relatively lower precision of 62%, leading to an F1 score of 69%. This suggests that while the model successfully identified most instances of conscientiousness, there were some false positives where the model misclassified other traits as conscientiousness. For agreeableness and Openness, which had a moderate performance, with an F1 score of 64% and 56%. The results indicate that the model struggles slightly with both precision and recall for these 2 classes, possibly due to overlap in features with other personality traits.

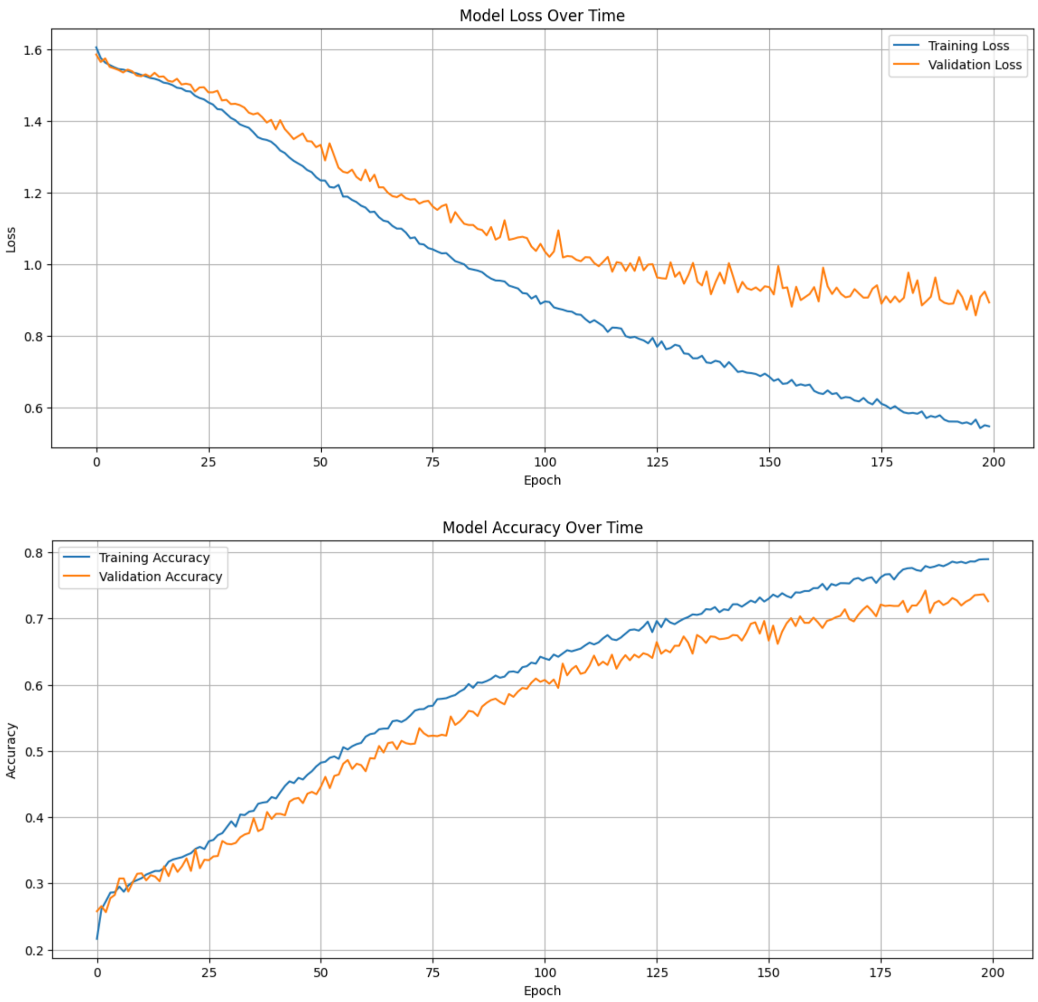


Figure 4.2.1.2 loss and accuracy curve of RNN

Figure 4.2.1.2 shows the loss and accuracy of the RNN model over 200 epochs. Both training loss and validation loss decrease steadily over time, indicating that the model is learning effectively. However, a noticeable gap emerges between the training and validation loss curves over time. While the training loss continues to decrease, the validation loss stabilizes, leading to divergence. This gap is a common indicator of overfitting, as the model becomes increasingly specialized for the training data while failing to generalize to unseen data. The small fluctuations in the validation loss may be attributed to Small batch sizes or noisy data in the validation set and also the complexity of the data, which might require additional regularization techniques to improve the stability.

4.2.2 XGBoost

Evaluating XGBoost on Testing Data...

	precision	recall	f1-score	support
agreeableness	0.79	0.77	0.78	708
conscientiousness	0.80	0.77	0.78	713
extraversion	0.78	0.95	0.86	726
neuroticism	0.87	0.85	0.86	697
openness	0.84	0.72	0.78	692
accuracy			0.81	3536
macro avg	0.82	0.81	0.81	3536
weighted avg	0.82	0.81	0.81	3536

Figure 4.2.2.1 classification report of XGBoost

```
Average Accuracy: 0.6316
Average Precision: 0.6285
Average Recall: 0.6316
Average F1 Score: 0.6226
```

Figure 4.2.2.2 KFold result of XGBoost

XGBoost delivers a respectable test accuracy of 81.33% but underperforms during cross-validation with 63.16% average accuracy. The discrepancy test metrics suggest possible overfitting to the test set. XGBoost delivers a weighted Precision of 85.13%, a weighted Recall of 81.33%, and a weighted F1 Score of 81.17%

The model performs well for agreeableness class, but a slightly lower recall of 78% suggests that a small portion of actual "agreeableness" instances are misclassified. Conscientiousness trait exhibits a balanced performance, though recall of 77% remains a limiting factor, indicating that the model could still improve in identifying some conscientiousness cases. Extraversion has the highest recall of 95%, meaning the model identifies most "extraversion" instances correctly. However, the precision is slightly lower, suggesting the model sometimes classifies other traits as "extraversion" incorrectly. This could be the same problem with the RNN model that the model is influenced by the upsampling technique, which increases the frequency of extraversion samples in the training data. The model performs exceptionally well for neuroticism, achieving one of the highest f1-scores of 86%. Both precision and recall are strong, indicating reliable predictions for this trait. Openness has a lower recall compared to its precision, meaning a significant number of "openness" instances are not correctly identified. This may suggest that the feature representation for openness is less distinct compared to other traits.

4.2.3 Random Forest

	precision	recall	f1-score	support
agreeableness	0.89	0.90	0.90	708
conscientiousness	0.92	0.93	0.93	713
extraversion	0.95	0.98	0.96	726
neuroticism	0.96	0.94	0.95	697
openness	0.93	0.90	0.91	692
accuracy			0.93	3536
macro avg	0.93	0.93	0.93	3536
weighted avg	0.93	0.93	0.93	3536

Figure 4.2.3.1 classification report of Random Forest

```
Average Accuracy: 0.7544
Average Precision: 0.7507
Average Recall: 0.7544
Average F1 Score: 0.7508
```

Figure 4.2.3.2 KFold result of Random Forest

Random Forest achieves 93% accuracy on the testing data, precision, recall, and f1-scores for all traits are consistently above 90%, demonstrating the robustness of the model in identifying all personality traits. However, despite excellent performance on the testing data, the k-fold cross-validation scores tell a different story with the accuracy being only 75.44% and the average f1-score of 75.08%. This substantial drop in performance during cross-validation suggests that the model is overfitting during training. This occurs because Random Forest may have learned patterns specific to the training data but struggles to generalize when evaluated across multiple folds.

The Random Forest model performs exceptionally well across all personality traits, with high precision and recall scores, particularly for extraversion of 95% precision score and recall score of 98% and neuroticism with precision score of 96% and recall score of 94%. This could be the same problem with the RNN model and Random Forest that the model is influenced by the upsampling technique, which increases the frequency of extraversion and neuroticism samples in the training data. While agreeableness and conscientiousness also achieve balanced and strong results, the slightly lower recall of 90% for openness suggests some difficulty in clearly separating its features from other traits.

4.2.4 SVM

	precision	recall	f1-score	support
agreeableness	0.77	0.76	0.76	708
conscientiousness	0.79	0.80	0.79	713
extraversion	0.87	0.93	0.90	726
neuroticism	0.82	0.80	0.81	697
openness	0.79	0.74	0.76	692
accuracy			0.81	3536
macro avg	0.81	0.81	0.81	3536
weighted avg	0.81	0.81	0.81	3536

Figure 4.2.4.1 classification report of SVM

```
Average Accuracy: 0.6313
Average Precision: 0.6253
Average Recall: 0.6313
Average F1 Score: 0.6262
```

Figure 4.2.4.2 KFold result of SVM

The SVM achieves a respectable test accuracy of 80.68% on the test data, precision, recall, and F1-score are all consistent, reflecting a balanced model with no significant bias towards any class. However, the average accuracy during k-fold cross-validation drops to 63.13%. This significant difference suggests the model might not generalize well across folds, potentially indicating some level of overfitting on the training set. The drop in precision and F1-score indicates that the model might struggle with certain personality trait classes during cross-validation.

The SVM model demonstrates moderate performance, excelling in identifying extraversion with precision score of 87% and recall score of 93%. But struggling slightly with agreeableness 77% precision score and 76% recall score and openness of 79% precision score and 74% recall due to less distinct feature separability and sensitivity to overlapping patterns.

4.2.5 Overall

	Accuracy	Precision	Recall	F1-Score
RNN	74.21	73.51	74.21	73.51
K-Fold RNN	71.37	70.74	71.37	70.59
XGBoost	65.53	65.23	65.53	64.69
K-Fold XGBoost	63.16	62.85	63.16	62.26
Random Forest	93.10	93.11	93.10	93.09
K-Fold RF	75.44	75.07	75.44	75.08
SVM	80.68	80.58	80.68	80.59
K-Fold SVM	63.13	62.53	63.13	62.62

The performance results from different models RNN, XGBoost, Random Forest, and SVM highlight varying levels of effectiveness in predicting personality traits based on the given dataset. While all models have demonstrated satisfactory performance in terms of accuracy, precision, recall, and F1 scores, several areas for improvement have emerged, particularly concerning overfitting and class imbalance. The models exhibited notable discrepancies between their performance on the test set and during k-fold cross-validation, suggesting that they may be overfitting to the training data and not generalizing well to unseen data.

While each model demonstrated relatively strong performance in predicting personality traits, they all showed signs of overfitting to the training data, as evidenced by the difference between test accuracy and k-fold cross-validation results. To mitigate overfitting, techniques such as regularization, cross-validation, hyperparameter tuning, feature engineering and ensemble methods should be prioritized. For class imbalance, methods like upsampling, SMOTE, and adjusting class weights should be considered, particularly for classes like extraversion and neuroticism that showed skewed results due to the preprocessing steps. Exploring the feature engineering and selection process will also help in improving model performance across all personality traits, ensuring better generalization to unseen data.

4.3 Overall Result and Discussion

The performance of both personality and emotion models reveals notable strengths but is marred by a significant issue which is overfitting. This challenge, observed across both models, impacts their ability to generalize effectively to new data, limiting their real-world applicability. For the emotion model, while algorithms like Random Forest and XGBoost achieved high test accuracy (above 78%), the sharp drop in k-fold cross-validation results to around 60% highlights their reliance on specific patterns in the training data. Additionally, the model struggled to accurately identify underrepresented emotions such as disgust and fear, likely due to overlapping feature representations and class imbalances. The angry class performed well due to its distinctive features, but the inconsistencies in handling rare emotions emphasize the need for better class balancing and data augmentation.

Similarly, the personality model exhibited overfitting, as evidenced by a significant gap between training and validation performance. Traits like extraversion were more accurately predicted, likely due to clearer behavioral cues, while subtler traits like agreeableness and openness posed challenges. The reliance on high-dimensional data, including all the audio features, combined with variability in data quality, further complicated the model's ability to generalize. Both models faced issues with noise and insufficient representation of nuanced features, making them less effective in diverse contexts.

To address these limitations, future work must incorporate robust regularization techniques such as dropout, early stopping, and L2 regularization to control model complexity. Data augmentation, especially for underrepresented classes can significantly enhance performance. Furthermore, techniques like ensemble learning and hyperparameter optimization can stabilize performance and reduce overfitting. By focusing on these strategies and validating the models on diverse, unseen datasets, their generalization capabilities can be improved. Emphasizing domain-specific customization and incorporating explainable AI will further enhance the practical utility and interpretability of these models, ensuring their relevance for real-world applications.

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